

HW/Lab 5 Sol: MIMO Diversity

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Question 1 Sol: 1×2 Receive Diversity with MRC

Given.

$$y_1 = h_1 x + n_1, \quad y_2 = h_2 x + n_2, \quad n_1, n_2 \sim \mathcal{CN}(0, N_0), \quad \mathbb{E}[|x|^2] = P.$$

(a) **Vector model.**

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}, \quad \mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}, \quad \Rightarrow \quad \mathbf{y} = \mathbf{h}x + \mathbf{n}.$$

Noise: $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_2)$, so $\mathbb{E}[\mathbf{n}\mathbf{n}^H] = N_0 \mathbf{I}_2$.

(b) **MRC decision statistic.** For linear combining $\hat{x} = \mathbf{w}^H \mathbf{y}$ with white noise, the SNR-maximizing weights satisfy $\mathbf{w} \propto \mathbf{h}$. A standard choice is $\mathbf{w} = \mathbf{h}$, yielding

$$\hat{x} = \mathbf{h}^H \mathbf{y} = h_1^* y_1 + h_2^* y_2.$$

(c) **Post-combiner SNR.** With $\mathbf{w} = \mathbf{h}$,

$$\hat{x} = \underbrace{\|\mathbf{h}\|_2^2}_{|h_1|^2 + |h_2|^2} x + \mathbf{h}^H \mathbf{n},$$

so the signal power is $P \|\mathbf{h}\|_2^4$ and the noise power is $N_0 \|\mathbf{h}\|_2^2$. Hence

$$\gamma_{\text{MRC}} = \frac{P \|\mathbf{h}\|_2^2}{N_0} = \frac{(|h_1|^2 + |h_2|^2)P}{N_0}.$$

(d) **Numerical example.** $h_1 = 1 + j \Rightarrow |h_1|^2 = 1^2 + 1^2 = 2$. $h_2 = 0.5 - j \Rightarrow |h_2|^2 = 0.5^2 + (-1)^2 = 0.25 + 1 = 1.25$. With $P = 1$, $N_0 = 0.5$:

$$\gamma_{\text{MRC}} = \frac{(2 + 1.25) \cdot 1}{0.5} = \frac{3.25}{0.5} = 6.5.$$

Single-branch SNRs:

$$\gamma_1 = \frac{2}{0.5} = 4, \quad \gamma_2 = \frac{1.25}{0.5} = 2.5.$$

(e) **Selection combining.**

$$\gamma_{\text{SC}} = \max\{\gamma_1, \gamma_2\} = \max\{4, 2.5\} = 4.$$

MRC-over-SC gain (dB):

$$10 \log_{10} \left(\frac{\gamma_{\text{MRC}}}{\gamma_{\text{SC}}} \right) = 10 \log_{10}(6.5/4) = 10 \log_{10}(1.625) \approx 2.11 \text{ dB}.$$

General ordering: since $\max\{a, b\} \leq a + b$, we have $\gamma_{\text{SC}} \leq \gamma_1 + \gamma_2 = \gamma_{\text{MRC}}$, with equality iff $\min\{\gamma_1, \gamma_2\} = 0$ (i.e., one branch contributes no SNR).

(f) **Quick BER (BPSK).** For coherent BPSK with instantaneous SNR γ :

$$P_b = Q(\sqrt{2\gamma}).$$

Thus, using the values above,

$$P_b^{(\text{MRC})} = Q(\sqrt{2 \cdot 6.5}) = Q(\sqrt{13}), \quad P_b^{(\text{SC})} = Q(\sqrt{2 \cdot 4}) = Q(2\sqrt{2}).$$

(Leaving answers in $Q(\cdot)$ form as requested.)

Question 2 Sol: 2×1 Transmit Diversity - CSIT Beamforming vs. Alamouti

Given. Channel $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$, noise $n \sim \mathcal{CN}(0, N_0)$, symbol power $\mathbb{E}[|x|^2] = P$, total TX power per time slot is P .

(a) **CSIT beamforming (derivation).**

(i) Received sample: $y = \mathbf{h}^\top \mathbf{w} x + n$ with $\|\mathbf{w}\|_2 = 1$. The instantaneous SNR is

$$\gamma = \frac{|\mathbf{h}^\top \mathbf{w}|^2 P}{N_0} \leq \frac{\|\mathbf{h}\|_2^2 P}{N_0},$$

with equality by Cauchy-Schwarz when $\mathbf{w} = \frac{\mathbf{h}^*}{\|\mathbf{h}\|_2}$ (channel-conjugate beamforming).

(ii) Thus $\gamma_{\text{BF}} = \frac{\|\mathbf{h}\|_2^2 P}{N_0} = \frac{(|h_1|^2 + |h_2|^2) P}{N_0}$.

(b) **"naïve repeat" (no CSIT) can destructively add.**

(i) With $\mathbf{x} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} x$,

$$y = \frac{h_1 + h_2}{\sqrt{2}} x + n, \quad \gamma_{\text{rep}} = \frac{|h_1 + h_2|^2}{2} \frac{P}{N_0}.$$

(ii) If $h_2 \approx -h_1$ (nearly opposite phase and similar magnitudes), then $|h_1 + h_2| \ll \|\mathbf{h}\|_2$, so $\gamma_{\text{rep}} \ll \gamma_{\text{BF}}$.

- (c) **Alamouti STBC (no CSIT).** Over two slots, transmit $[x_1 \ x_2]$ then $[-x_2^* \ x_1^*]$ (total power P per slot).

(i) Received samples:

$$y_t = h_1 x_1 + h_2 x_2 + n_t, \quad y_{t+1} = -h_1 x_2^* + h_2 x_1^* + n_{t+1}.$$

(ii) The standard combiner

$$\tilde{x}_1 = h_1^* y_t + h_2 y_{t+1}^*, \quad \tilde{x}_2 = h_2^* y_t - h_1 y_{t+1}^*$$

yields

$$\tilde{x}_1 = (|h_1|^2 + |h_2|^2) x_1 + \underbrace{h_1^* n_t + h_2 n_{t+1}^*}_{\tilde{n}_1}, \quad \tilde{x}_2 = (|h_1|^2 + |h_2|^2) x_2 + \underbrace{h_2^* n_t - h_1 n_{t+1}^*}_{\tilde{n}_2}.$$

Thus each symbol decouples with effective gain $|h_1|^2 + |h_2|^2$.

(iii) Since $\mathbb{E}[\tilde{n}_i^2] = N_0(|h_1|^2 + |h_2|^2)$ and $\mathbb{E}[|x_1|^2] = \mathbb{E}[|x_2|^2] = \frac{P}{2}$, the per-symbol SNR is

$$\gamma_{\text{Alam}} = \frac{(|h_1|^2 + |h_2|^2) P}{2N_0},$$

with code rate 1 (two slots transmit two symbols).

- (d) **Numerical comparison.** Let $h_1 = 0.6 + 0.8j$ ($|h_1|^2 = 1$), $h_2 = -0.3 + 0.4j$ ($|h_2|^2 = 0.25$), hence $\|\mathbf{h}\|_2^2 = 1.25$, with $P = 1$, $N_0 = 0.2$.

(i) Already computed: $|h_1|^2 = 1$, $|h_2|^2 = 0.25$, $\|\mathbf{h}\|_2^2 = 1.25$.

(ii) **CSIT beamforming:**

$$\gamma_{\text{BF}} = \frac{1.25}{0.2} = 6.25 \quad \Rightarrow \quad 10 \log_{10}(6.25) \approx 7.96 \text{ dB}.$$

(iii) **naïve repeat:** $h_1 + h_2 = 0.3 + 1.2j$, so $|h_1 + h_2|^2 = 0.09 + 1.44 = 1.53$,

$$\gamma_{\text{rep}} = \frac{1.53}{2} \cdot \frac{1}{0.2} = 0.765 \cdot 5 = 3.825 \quad \Rightarrow \quad 10 \log_{10}(3.825) \approx 5.82 \text{ dB}.$$

Loss vs BF: $7.96 - 5.82 \approx 2.14$ dB (equivalently $10 \log_{10}(6.25/3.825) \approx 2.13$ dB).

(iv) **Alamouti:**

$$\gamma_{\text{Alam}} = \frac{1.25}{2 \cdot 0.2} = 3.125 \quad \Rightarrow \quad 4.95 \text{ dB},$$

which is around 3 dB lower than CSIT beamforming.

- (e) **Short design note.** Alamouti's orthogonal structure avoids self-interference and guarantees an effective gain of $|h_1|^2 + |h_2|^2$ without CSIT, whereas naïve repetition can suffer destructive addition when h_1 and h_2 are out of phase, leading to significantly lower SNR at the same total transmit power.

Question 3 Sol: Simple 2×2 MIMO with Zero-Forcing

(a) **ZF equalizer and effective noise.** With $\mathbf{W} = \mathbf{H}^{-1}$,

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{y} = \mathbf{H}^{-1}(\mathbf{H}\mathbf{x} + \mathbf{n}) = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}.$$

Let the rows of $\mathbf{H}^{-1}$ be $\mathbf{w}_1^H$ and $\mathbf{w}_2^H$. Then the decision statistics are

$$\hat{x}_1 = \mathbf{w}_1^H \mathbf{y} = x_1 + \underbrace{\mathbf{w}_1^H \mathbf{n}}_{\triangleq n_{\text{eff},1}}, \quad \hat{x}_2 = \mathbf{w}_2^H \mathbf{y} = x_2 + \underbrace{\mathbf{w}_2^H \mathbf{n}}_{\triangleq n_{\text{eff},2}}.$$

Since $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_2)$, the effective noises are

$$n_{\text{eff},i} \sim \mathcal{CN}(0, N_0 \|\mathbf{w}_i\|^2), \quad i \in \{1, 2\}.$$

(b) **Numeric evaluation.** Given

$$\mathbf{H} = \begin{bmatrix} 1 & 0.5 \\ -0.3 & 1.2 \end{bmatrix}, \quad P = 1, \quad N_0 = 0.2,$$

(i) *Inverse $\mathbf{H}^{-1}$.* For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Here $\det(\mathbf{H}) = 1 \cdot 1.2 - 0.5(-0.3) = 1.35$, so

$$\mathbf{H}^{-1} = \frac{1}{1.35} \begin{bmatrix} 1.2 & -0.5 \\ 0.3 & 1 \end{bmatrix} = \begin{bmatrix} \frac{8}{9} & -\frac{10}{27} \\ \frac{2}{9} & \frac{20}{27} \end{bmatrix} \approx \begin{bmatrix} 0.8889 & -0.3704 \\ 0.2222 & 0.7407 \end{bmatrix}.$$

(ii) *Squared row norms.* With $\mathbf{w}_1 = [\frac{8}{9}, -\frac{10}{27}]$ and $\mathbf{w}_2 = [\frac{2}{9}, \frac{20}{27}]$,

$$\|\mathbf{w}_1\|^2 = \left(\frac{8}{9}\right)^2 + \left(-\frac{10}{27}\right)^2 = \frac{64}{81} + \frac{100}{729} = \frac{676}{729} \approx 0.9273,$$

$$\|\mathbf{w}_2\|^2 = \left(\frac{2}{9}\right)^2 + \left(\frac{20}{27}\right)^2 = \frac{4}{81} + \frac{400}{729} = \frac{436}{729} \approx 0.5981.$$

(iii) *Post-ZF SNRs.* Using $\gamma_i = \frac{P}{N_0 \|\mathbf{w}_i\|^2}$ with $P = 1$ and $N_0 = 0.2$,

$$\gamma_1 = \frac{1}{0.2} \cdot \frac{729}{676} = \frac{729}{135.2} \approx 5.392, \quad \gamma_2 = \frac{1}{0.2} \cdot \frac{729}{436} = \frac{729}{87.2} \approx 8.360.$$

(iv) *More noise-enhanced stream and SNR ratio.* Stream 1 has the *larger* row norm (more noise enhancement) and thus the lower SNR. The SNR ratio is

$$\frac{\gamma_1}{\gamma_2} = \frac{\|\mathbf{w}_2\|^2}{\|\mathbf{w}_1\|^2} = \frac{436}{676} = \frac{109}{169} \approx 0.645 \Rightarrow 10 \log_{10}\left(\frac{\gamma_1}{\gamma_2}\right) \approx -1.90 \text{ dB}.$$

(c) **Explanation (when ZF degrades).** ZF suffers severe noise enhancement when $\mathbf{H}$ is *ill-conditioned*/near-singular, i.e.,

$$\kappa(\mathbf{H}) = \frac{\sigma_{\max}(\mathbf{H})}{\sigma_{\min}(\mathbf{H})} \gg 1 \quad (\text{equivalently, } \sigma_{\min}(\mathbf{H}) \approx 0, \text{ or } |\det(\mathbf{H})| \text{ small, or columns nearly collinear}).$$

In that case $(\mathbf{H}^H \mathbf{H})^{-1}$ has large diagonal entries, inflating the post-ZF noise variances $N_0 \|\mathbf{w}_i\|^2$ and driving the SNRs down.

Question 4 Sol: MIMO Throughput for an 802.11n-like Link

Given. $P = 10$, $N_0 = 0.1$. Single scenario with $h = h_1 = 0.6 + 0.2j$, $h_2 = 0.4 - 0.8j$. SNR (dB):

$$\gamma_{\text{SISO,dB}} = 10 \log_{10} \left(\frac{|h|^2 P}{N_0} \right), \quad \gamma_{\text{MRC,dB}} = 10 \log_{10} \left(\frac{(|h_1|^2 + |h_2|^2) P}{N_0} \right).$$

MCS thresholds (dB): $\{3, 6, 10, 14, 18, 22, 26, 30, 34\}$ for BPSK $\frac{1}{2}$, QPSK $\frac{1}{2}$, QPSK $\frac{3}{4}$, 16-QAM $\frac{1}{2}$, 16-QAM $\frac{3}{4}$, 64-QAM $\frac{2}{3}$, 64-QAM $\frac{3}{4}$, 256-QAM $\frac{2}{3}$, 256-QAM $\frac{3}{4}$. Bits/OFDM symbol (single stream): $52 b_{\text{mod}} R$ with $b_{\text{mod}} \in \{1, 2, 4, 6, 8\}$. Throughput (Mbps): (bits/symbol) $\times 250,000/10^6$.

Channel magnitudes.

$$|h|^2 = |h_1|^2 = 0.6^2 + 0.2^2 = 0.40, \quad |h_2|^2 = 0.4^2 + (-0.8)^2 = 0.16 + 0.64 = 0.80, \quad |h_1|^2 + |h_2|^2 = 1.20.$$

(a) **Compute SNRs (dB) and choose MCS.**

$$\gamma_{\text{SISO}} = \frac{0.40 \cdot 10}{0.1} = 40 \Rightarrow \gamma_{\text{SISO,dB}} = 10 \log_{10}(40) = \boxed{16.02 \text{ dB}},$$

$$\gamma_{\text{MRC}} = \frac{1.20 \cdot 10}{0.1} = 120 \Rightarrow \gamma_{\text{MRC,dB}} = 10 \log_{10}(120) = \boxed{20.79 \text{ dB}}.$$

MCS selections (highest threshold $\leq$ SNR in dB):

$$\text{SISO (16.02 dB)} \Rightarrow \boxed{16\text{-QAM, } R = \frac{1}{2}} \quad (14 \leq 16.02 < 18),$$

$$\text{MRC (20.79 dB)} \Rightarrow \boxed{16\text{-QAM, } R = \frac{3}{4}} \quad (18 \leq 20.79 < 22).$$

(b) **Throughput and improvement ratio.**

$$\text{SISO: bits/symbol} = 52 \times 4 \times \frac{1}{2} = \boxed{104} \Rightarrow \text{throughput} = \frac{104 \times 250,000}{10^6} = \boxed{26.0 \text{ Mbps}},$$

$$\text{MRC: bits/symbol} = 52 \times 4 \times \frac{3}{4} = \boxed{156} \Rightarrow \text{throughput} = \frac{156 \times 250,000}{10^6} = \boxed{39.0 \text{ Mbps}}.$$

$$\text{Improvement ratio} = \boxed{39.0/26.0 = 1.5 \times}.$$

(c) **Explanation (short).** MRC combines two receive branches, raising effective gain from $|h|^2 = 0.40$ to $|h_1|^2 + |h_2|^2 = 1.20$, which in dB moves the link from 16.02 dB to 20.79 dB. This crosses the 18 dB threshold, allowing a higher code rate (16-QAM $\frac{3}{4}$) than SISO (16-QAM $\frac{1}{2}$), hence $1.5 \times$ higher throughput.

(d) *If the MCS is chosen wrongly:*

- **Too conservative (choose a lower MCS than allowed):** Throughput *decreases* (fewer bits per symbol), BER *decreases* (more robust than necessary).
- **Too aggressive (choose a higher MCS than allowed):** Nominal bits/symbol are higher, but actual packet/bit errors *increase sharply*; effective throughput *decreases* due to errors/retransmissions, and BER *increases*.

Question 5 Sol: Programming: Rank, Invertibility, and SVD (2×2)

```
=== H1 ===
```

```
    1    1
    1    1
```

- (a) Singular values ($\sigma_1 \geq \sigma_2$): [2.0000, 0.0000]
 rank(H) = 1, det(H) = 0.000000 --> Invertible: false
 (b) ZF not applicable (non-invertible; $\sigma_{\min} = 0$).

```
=== H2 ===
```

```
    1.0000    0.6000
   -0.2000    1.1000
```

- (a) Singular values ($\sigma_1 \geq \sigma_2$): [1.3298, 0.9175]
 rank(H) = 2, det(H) = 1.220000 --> Invertible: true
 (b) $\kappa_2 = \sigma_{\max}/\sigma_{\min} = 1.4494$
 Classification: WELL-CONDITIONED (ZF noise amplification is low).

```
=== H3 ===
```

```
    0    1.0000
    1.0000    0.1000
```

- (a) Singular values ($\sigma_1 \geq \sigma_2$): [1.0512, 0.9512]
 rank(H) = 2, det(H) = -1.000000 --> Invertible: true
 (b) $\kappa_2 = \sigma_{\max}/\sigma_{\min} = 1.1051$
 Classification: WELL-CONDITIONED (ZF noise amplification is low).
 (c) When $\sigma_{\min} \rightarrow 0$ (near-singular H), choose SINGLE-STREAM DIVERSITY
 (TX diversity / RX MRC) at the same total power instead of ZF spatial multiplexing,
 because ZF amplifies noise by $\sim 1/\sigma_{\min}$ on the weak singular mode.

```
1 %% Question 5 Solution: Rank, Invertibility, and SVD (2x2)
2 % Computes singular values, rank, determinant, invertibility, and 2-
  norm
3 % condition number for each channel; classifies ZF sensitivity; and
  includes
4 % one-sentence explanations as requested.
5
6 clear; clc;
7
8 % —— Channels ——
9 H = {
10     [1 1; 1 1], ...
11     [1 0.6; -0.2 1.1], ...
12     [0 1; 1 0.1]
13 };
14 names = ["H1", "H2", "H3"];
```

```

15
16 fprintf('== Question 5: MATLAB Solution ==\n');
17 fprintf('SVD-based conditioning governs ZF noise amplification (
    singular values, not eig(H)).\n');
18
19 for k = 1:numel(H)
20     A = H{k};
21     [U,S,V] = svd(A);           % singular values (descending on
    diagonal)
22     sig = diag(S).';           % [sigma1 sigma2], sigma1 >= sigma2 >=
    0
23     rnk = rank(A);
24     d = det(A);
25     kappa2 = cond(A,2);        % 2-norm condition number = sigma_max/
    sigma_min
26
27     fprintf('\n== %s ==\n', names(k));
28     disp(A);
29
30     % (a) Singular values, rank, determinant, invertibility
31     fprintf('(a) Singular values (sigma1>=sigma2): [%4f, %4f]\n', sig
    (1), sig(2));
32     fprintf('    rank(H) = %d, det(H) = %6f  ->  Invertible: %s\n',
    ...
33         rnk, d, string(abs(d) > 0));
34
35     % (b) Condition number & ZF sensitivity (only if invertible)
36     if abs(d) > 0
37         fprintf('(b) kappa2 = sigma_max/sigma_min = %4f\n', kappa2);
38         if kappa2 < 3
39             fprintf('    Classification: WELL-CONDITIONED (ZF noise
    amplification is low).\n');
40         elseif kappa2 < 10
41             fprintf('    Classification: MODERATE (ZF may modestly amplify
    noise on weak mode).\n');
42         else
43             fprintf('    Classification: POORLY-CONDITIONED (ZF strongly
    amplifies noise).\n');
44         end
45     else
46         fprintf('(b) ZF not applicable (non-invertible; sigma_min = 0).\n')
    ;
47     end
48 end
49
50 % (c) One-sentence explanation (printed once)
51 fprintf(['\n(c) When sigma_min -> 0 (near-singular H), choose SINGLE-
    STREAM DIVERSITY\n' ...

```

```

52         '      (TX diversity / RX MRC) at the same total power instead
           of ZF spatial multiplexing,\n' ...
53         '      because ZF amplifies noise by  $\sim 1/\sigma_{\min}$  on the weak
           singular mode.\n']]);
54
55 %% Optional (for your PDF): Brief SVD diagonalization statement
56 fprintf(['\nSVD diagonalization:  $y = H x + n = U \Sigma V^H x + n$ .\n'
           ...
57         'Premultiply  $U^H$ :  $y_{\text{tilde}} = U^H y = \Sigma x_{\text{tilde}} + n_{\text{tilde}}$ ,
           with  $x_{\text{tilde}} = V^H x$ .\n' ...
58         'Because  $U$  is unitary,  $\text{cov}(n_{\text{tilde}}) = N_0 I$  (noise statistics
           preserved).\n']);

```